

3. Domain Decomposition

To make the EcoGrid Energy platform maintainable, scalable and evolvable, the system architecture should be scrupulously broken down into the components of Domain-Driven Design (DDD) principles. The main bottleneck will be to ensure that the high-frequency characteristic of the IoT telemetry is not a source of poor performance and performance of the main business activity. Thus, the system is divided into three different bounded contexts [1].

3.1 Bounded Contexts

A Bounded Context is a logical boundary beyond which a particular domain model is said to be valid and consistent [1, 2]. In the case of EcoGrid, we have detected the following contexts:

- **Smart Meter Integration Context (IoT Telemetry):**
 - Scope: It is a context specific to the real-time ingestion, validation, and aggregation of energy generation and consumption data of IoT devices and smart meters.
 - Justification: Decentralisation of peer-to-peer (P2P) platforms significantly depends on the smart meters as secure nodes [3]. The amount of information produced by these meters (e.g. minute-by-minute wattage) is huge. We have confined the high-velocity data by isolating this domain. The main task here is to process raw time-series data into practical, aggregated, energy intervals (e.g., hourly net generation) by not overloading the transactional systems.
- **Marketplace Context (P2P Local Energy Market):**
 - Scope: It is the central business area that will be tasked to match prosumers (residential sellers) with consumers (buyers). It maintains user accounts, energy trading preferences, ongoing bids, and matching engine algorithm.
 - Justification: Local energy markets that are successful need to trade off individual preferences of peers against the general effectiveness [3]. Since this domain is important to transactional integrity and complex matching algorithms, not to mention high-frequency data streams, decoupling it and the Smart Meter Integration means that the responsiveness of UI and trading logic is not constrained by IoT ingestion spikes.
- **Financial Settlement Context (Micro-transactions & Clearing):**
 - Scope: This context does the secure micro-transactions of electricity trades. It controls

digital wallets, unchangeable entries of ledgers, billing cycles and integrations with external payment choices.

- Justification: Domain-driven design proves to be extremely useful in organizing financial architecture around core domains to enable systems to scale and manage [4]. The field of the highest accuracy, safety, and auditing is required. It should be separated so that financial calculations (involving determining network utilization charges basing on grid topology [3]) would be totally resistant to external system breakdowns.